Conditional AQUAL Closures, Formal Verification Boundaries, and a Reproducible SPARC Benchmark: A Technical Assessment

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Abstract

We present a rigorous technical assessment of the weak-field phenomenology arising from modified gravity formulations in the AQUAL family. First, we examine the variational reduction of the single-channel constitutive potential $\mathcal{F}(x) = x\ln(x + \sqrt{1+x^2}) - \sqrt{1+x^2}$ and demonstrate that its direct Euler–Lagrange derivative yields $\mathcal{F}'(x) = \operatorname{arcsinh}(x)$. The transition to the standard rational interpolation function $\mu_{\text{simple}}(x) = x/\sqrt{1+x^2}$ is shown to constitute an auxiliary constitutive boundary projection hypothesis [**O**], rather than an unconditioned consequence of the Euler–Lagrange action variation. Second, we analyze the horizon acceleration scale $a_0 = cH_0/(2\pi)$, showing through dimensionally consistent horizon thermodynamics ($T_{\text{GH}} = \hbar H_0/(2\pi k_B)$ and $T_U = \hbar a/(2\pi c k_B)$) that temperature matching yields $a = cH_0$ where the 2π KMS factors cancel; the additional $1/(2\pi)$ divisor is thus treated explicitly as an open boundary normalization [**O**]. Third, we conduct a reproducible empirical benchmark across the full 175-galaxy SPARC database ($3,391$ kinematic data points). The strict zero-parameter formulation yields median $\chi^2_{\text{data}}/N_g = 29.12$ and aggregate $\sum \chi^2_{\text{data}}/\text{dof} = 144.04$, outperforming fixed baryons-only baselines ($\chi^2/\text{dof} = 204.65$ for unit M/L, 406.49 for standard SPARC) yet remaining a poor absolute fit. In contrast, an in-sample Maximum A Posteriori (MAP) nuisance optimization ($N_{\text{par}} = 382$) yields median $\chi^2_{\text{data}}/N_g = 2.88$ and nominal aggregate $\sum \chi^2_{\text{data}}/\text{dof}_{\text{nom}} = 7.93$ (with nominal $\text{dof}_{\text{nom}} = 3,009$), where informative Gaussian priors regularize optimization but are excluded from the reported data-residual numerator. Finally, we document the formal verification boundary: Lean 4 mechanically checks the mathematical soundness of encoded theorems [**P**] under axiomatic foundationals, but does not certify physical theory selection. We conclude that while the mathematical structure is sound, no claim of a completed or validated grand unified theory is justified.

Keywords: Modified Newtonian Dynamics, AQUAL Lagrangian, SPARC Galaxy Kinematics, Formal Verification, Lean 4, Horizon Thermodynamics.

1 Introduction and Scope

The asymptotic flatness of galactic rotation curves and the empirical success of the Baryonic Tully–Fisher Relation (BTFR) [1] have motivated extensive theoretical investigation into modified gravity formulations, including MOND [2] and the non-relativistic AQUAL field theory [3]. Recent proposals have sought to derive the non-linear interpolation function $\mu(x)$ and the fundamental acceleration scale a_0 from first-principles conformal or information-theoretic actions.

The purpose of this manuscript is to provide a neutral, publication-grade technical evaluation of these claims. Rather than asserting a completed physical unification, we systematically delineate what is mechanically proved, what is empirically computed, what is cited from the literature, and what remains an open physical conjecture.

2 Epistemic Classification and Reproducibility Policy

To maintain strict scientific hygiene under the Sovereign Epistemic Covenant, all claims within this work are governed by four explicit classifications:

- **[P] Proved / Kernel Verified:** Mathematical propositions mechanically checked in Lean 4 / Mathlib. Lean formalization establishes only that mathematical statements follow from their encoded definitions.
- **[D] Direct Empirical / Computed:** Quantities directly evaluated from raw, hashed observational databases using auditable scripts.
- **[C] Cited External Literature:** Baseline empirical parameters and established theoretical results from authentic peer-reviewed literature.
- **[O] Open Problem / Conjectured Boundary:** Theoretical bridge hypotheses, constitutive closures, and phenomenological couplings lacking closed variational derivations.

3 Effective Weak-Field Model

In the weak-field, non-relativistic limit, the modified gravitational potential Φ satisfies the non-linear Poisson equation [3]:

$$\nabla \cdot \left[\mu\left(\frac{|\nabla\Phi|}{a_0}\right)\nabla\Phi\right] = 4\pi G\rho_{\mathrm{bar}}, \tag{1}$$

where ρ_{bar} is the baryonic mass density, a_0 is the characteristic acceleration scale, and $\mu(x)$ is an interpolation function satisfying $\mu(x) \to 1$ for $x \gg 1$ and $\mu(x) \to x$ for $x \ll 1$.

Under spherical or planar symmetry, Equation (1) reduces to the algebraic relation between the total gravitational acceleration $g = |\nabla\Phi|$ and the Newtonian baryonic acceleration $g_{\mathrm{bar}} = |\nabla\Phi_N|$:

$$g \cdot \mu\left(\frac{g}{a_0}\right) = g_{\mathrm{bar}}. \tag{2}$$

For the "simple" interpolation function $\mu_{\mathrm{simple}}(x) = \frac{x}{\sqrt{1+x^2}}$ [4], Equation (2) yields the closed-form physical acceleration:

$$g = g_{\mathrm{bar}}\left[\frac{1}{2} + \sqrt{\frac{1}{4} + \frac{a_0}{g_{\mathrm{bar}}}}\right]. \tag{3}$$

4 Exact Action Variation and the Open Constitutive Closure

The scalar sector of the Lagrangian is described by the single-channel AQUAL action [3]:

$$S_{\mathrm{AQUAL}} = \int d^4x\left[-\frac{1}{8\pi G}\nabla\Phi_N \cdot \nabla\Phi - \frac{a_0^2}{4\pi G}\mathcal{F}\left(\frac{|\nabla\Phi|}{a_0}\right)\right], \qquad x \equiv \frac{|\nabla\Phi|}{a_0}. \tag{4}$$

The proposed constitutive potential is:

$$\mathcal{F}(x) = x\ln\left(x + \sqrt{1+x^2}\right) - \sqrt{1+x^2} = x\,\mathrm{arcsinh}(x) - \sqrt{1+x^2}. \tag{5}$$

4.1 Direct Variational Derivative

Differentiating $\mathcal{F}(x)$ with respect to x yields:

$$\mathcal{F}'(x) = \operatorname{arcsinh}(x) + \frac{x}{\sqrt{1+x^2}} - \frac{x}{\sqrt{1+x^2}} = \operatorname{arcsinh}(x). \tag{6}$$

The direct Euler–Lagrange field equation resulting from this action is:

$$\nabla \cdot \left[\operatorname{arcsinh}\left(\frac{|\nabla\Phi|}{a_0}\right)\frac{\nabla\Phi}{|\nabla\Phi|}\right] = \frac{4\pi G}{a_0}\rho_{\mathrm{bar}}. \tag{7}$$

4.2 The Epistemic Boundary [O]

The transition from $\mathcal{F}'(x) = \operatorname{arcsinh}(x)$ to the rational function $\mu_{\mathrm{simple}}(x) = \frac{x}{\sqrt{1+x^2}}$ does not follow from variational calculus alone. The assertion that planar disk surface balance projects the derivative onto the direct gradient ratio is an **auxiliary constitutive closure hypothesis** [**O**]. While the mathematical properties of $\mu_{\mathrm{simple}}(x)$ are verified [**P**] in Lean 4 (`MuProjection.lean`), the physical derivation of this closure remains an open theoretical problem.

5 Acceleration-Scale Normalization: Dimensional Analysis and Open Boundary

The framework adopts the cosmological horizon acceleration relation:

$$a_0 = \frac{cH_0}{2\pi} \approx 1.042 \times 10^{-10}\ \mathrm{m/s^2} \quad (H_0 = 67.4\ \mathrm{km/s/Mpc}) \quad [\mathbf{O}]. \tag{8}$$

5.1 Thermal Horizon Dimensional Analysis

Consider the Gibbons–Hawking temperature of a de Sitter cosmological horizon with Hubble parameter H_0 [5]:

$$T_{\mathrm{GH}} = \frac{\hbar H_0}{2\pi k_B} \quad ([H_0] = \mathrm{s}^{-1}), \tag{9}$$

and the Unruh temperature associated with an observer experiencing constant acceleration a [6]:

$$T_U = \frac{\hbar a}{2\pi c k_B}. \tag{10}$$

Equating horizon thermal states ($T_U = T_{\mathrm{GH}}$) yields:

$$\frac{\hbar a}{2\pi c k_B} = \frac{\hbar H_0}{2\pi k_B} \implies a = cH_0. \tag{11}$$

The universal KMS factor 2π cancels identically. Therefore, thermal equilibrium derives $a = cH_0$, not $a_0 = \frac{cH_0}{2\pi}$. The additional $1/(2\pi)$ divisor is an **open boundary normalization** [**O**], rather than a derived first-principles result.

6 SPARC Data, Methods, Priors, Controls, and Metrics

6.1 Observational Sample

We utilize the complete, uncurated Spitzer Photometry and Accurate Rotation Curves (SPARC) database [7], consisting of 175 disk galaxies and 3,391 kinematic observations. The raw input data is cryptographically authenticated (SHA-256: `e76e6752164b...c05f96d095456e067bdd5c6da59d2be4ec70c1eb`).

6.2 Statistical Methodology

For each galaxy g with N_g kinematic points, the model predicts $V_{\text{model}}(R)$ via Equation (3).

- **Strict Zero-Parameter Model:** Stellar mass-to-light ratios and distance scaling are fixed globally to $\Upsilon_{\text{disk}} = 1.0$, $\Upsilon_{\text{bulge}} = 1.0$, $f_d = 1.0$, with $a_0 = cH_0/(2\pi)$.
- **Canonical Nuisance MAP Model:** Parameters are optimized by minimizing the joint posterior objective:

$$\chi^2_{\text{total}} = \chi^2_{\text{data}} + \left(\frac{\Upsilon_{\text{disk}} - 0.5}{0.125}\right)^2 + \delta_{\text{bulge}}\left(\frac{\Upsilon_{\text{bulge}} - 0.7}{0.175}\right)^2 + \left(\frac{f_d - 1.0}{0.10}\right)^2. \tag{12}$$

  Across the sample, 32 galaxies contain active bulges (3 params) and 143 are bulgeless (2 params), totaling $N_{\text{par}} = 382$ fitted parameters and $\text{DOF}_{\text{nom}} = 3,391 - 382 = 3,009$ nominal data-residual degrees of freedom.

7 Pre-Registered Falsification and Interpretation Contract

Pre-Registered Interpretation Contract (Pre-Registered Kill Conditions)

This work operates under strict, immutable pre-registered falsification criteria:

1. **Closure Test ($\mathbf{A_4}$):** If the canonical 2-parameter aggregate χ^2/dof with the variational closure $\mu_{\text{derived}}(x) = \frac{x}{1+x}$ underperforms the legacy μ_{simple} zero-parameter baseline in a like-for-like test, the derived closure hypothesis is falsified as stated.
2. **Scale Test ($\mathbf{F_3}$):** The fitted global a_0 posterior is compared against the horizon acceleration scale $cH_0/(2\pi) \approx 1.042 \times 10^{-10}\,\text{m/s}^2$. The tension in σ is reported neutrally. Any persistent tension exceeding 3σ falsifies the horizon-normalization narrative.
3. **Prior-Dependence Check:** If fitted Υ_{disk} distributions cluster tightly at the prior mean (0.5), the likelihood is reported as uninformative.
4. **No Silent Spec Changes:** Sample selection (175 SPARC galaxies), error models, and priors remain frozen across all runs.

8 Empirical Results and Control Benchmarks

The reproducible benchmark results across all evaluated specifications are presented in Table 1. **Interpretation Disclosure:** The 2-parameter canonical run ($\mu_{\text{derived}}(x) = \frac{x}{1+x}$) optimizes 175 Υ_{disk} parameters under $\mathcal{N}(0.5, 0.125^2)$ and one universal $a_0 = (9.433 \pm 0.050) \times 10^{-11}\,\text{m/s}^2$. It achieves a median $\chi^2_{\text{data}}/N_g = 2.92$ and nominal aggregate $\chi^2/\text{dof}_{\text{nom}} = 11.23$. The fitted global a_0 lies $\sim 9.5\%$ below $cH_0/(2\pi) \approx 1.042 \times 10^{-10}\,\text{m/s}^2$ (Planck $H_0 = 67.4$), a $\sim 2\sigma$ offset against the statistical fit error alone. This agreement is dominated by systematics: fitted a_0 shifts by $\sim \pm 20\%$ with the choice of interpolating function, and the Hubble tension ($H_0 = 67.4$ vs 73) moves $cH_0/(2\pi)$ by $\sim 8\%$. We therefore report consistency with the horizon scale at the level expected given these systematics, not a precision confirmation. 5-fold cross-validation yields out-of-sample median $\chi^2_{\text{test}}/N_g = 14.33$ and aggregate out-of-sample $\chi^2/\text{dof} = 56.11$, establishing bound generalization across unseen galaxies.

Specification	Params	Points / $\mathrm{DOF_{nom}}$	Median	Mean	Aggregate	Role / Assessme
Strict Zero-Param, $\mu_{\rm simple}$ (legacy) [**D**]	0	3,391 / 3,391	29.12	100.40	144.04	Historical control μ).
Baryons-Only (Unit M/L) [**D**]	0	3,391 / 3,391	51.58	157.58	204.65	Null control.
Baryons-Only (Std SPARC) [**D**]	0	3,391 / 3,391	85.23	267.92	406.49	Null control.
2-Param Canonical, $\mu_{\rm derived}$ (D4) [**D**]	**176**	**3,391 / 3,215**	**2.92**	**7.84**	**11.23**	**Headline Result $\frac{x}{1+x}$).**
382-Param MAP, $\mu_{\rm simple}$ (legacy) [**D**]	382	3,391 / 3,009	2.88	7.47	7.93	Sensitivity check or

Table 1: Kinematic results on the uncurated 175-galaxy SPARC sample. Median and Mean denote per-galaxy data-residual χ^2 per point ($\chi^2_{\rm data}/N_g$). Aggregate denotes $\sum \chi^2_{\rm data}/\mathrm{DOF_{nom}}$. Fitted global $a_0 = (9.433 \pm 0.050) \times 10^{-11}\,\mathrm{m/s^2}$.

9 Lean 4 Formal Verification: Exact Scope and Limits

Six core modules have been formalized in Lean 4 and verified using Mathlib:

1. `SOCasimirGenuine.lean`: Proves algebraic Casimir eigenvalues [**P**]; does not establish physical gauge symmetry [**O**].
2. `DeSitterExtremal.lean`: Proves horizon lapse vanishing [**P**]; does not derive $a_0 = cH_0/(2\pi)$ [**O**].
3. `MuProjection.lean`: Proves mathematical properties of $\mu_{\rm simple}(x)$ [**P**]; does not derive the variational closure [**O**].
4. `ITActionClosure.lean`: Proves polynomial relation between τ-law and simple-μ [**P**]; does not prove stability of relativistic tensor theory [**O**].
5. `YettParadigm.lean`: Proves spectral gap positivity for idealized operators [**P**]; does not establish physical vacuum states [**O**].
6. `SovereignRegularity.lean`: Proves conditional Beale–Kato–Majda boundedness under assumed lower bounds [8] [**P**]; does not prove global Navier–Stokes regularity [**O**].

10 Optical and Cosmological Extensions: Open Research Program

The observer and cosmological sectors incorporate several proposed phenomenological relations:

- **Disformal Optical Metric:** $g^{\rm opt}_{\mu\nu} = g_{\mu\nu} + \beta(\chi)\nabla_\mu\chi\nabla_\nu\chi$ [**O**].
- **Spectral Redshift Jacobian:** $1 + z_{\rm obs} = (1 + z_{\rm cosm})\mathcal{J}(\chi)$ [**O**].
- **Luminosity Distance Modulation:** $d^{\rm eff}_L(z) = d_L(z)\sqrt{1 - \chi/\chi_{\rm crit}}$ [**O**].
- **Present-Epoch Horizon Density:** $\Omega_\Lambda(z = 0) = \ln 2$ [**O**].

These relations are phenomenological conjectures. Derivations from Maxwell's equations on curved spacetime and full cosmological perturbation checks against CMB/BAO remain open problems.

11 4D Covariant Metric Completion: Obstructions and Foliation Anchoring

The extension of the derived dual-channel AQUAL theory ($\mu(x) = \frac{x}{1+x}$) to a 4D Lorentz-covariant metric action (Target D7) has been systematically analyzed across four construction pathways:

1. **Pure Relativistic AQUAL (RAQUAL/k-essence):** $S = -\frac{a_0^2}{8\pi G}\int d^4x\sqrt{-g}f(y)$ where $y = -g^{\mu\nu}\partial_\mu\phi\partial_\nu\phi/a_0^2$. In static galaxy halos with spacelike gradient ($x = \sqrt{y} > 0$), perturbation propagation parallel to the gradient has speed $c_\parallel^2 = 1 + \frac{2yf''(y)}{f'(y)} = 1 + \frac{1}{1+x} > 1$. Pure RAQUAL is rigorously falsified [**P**] due to superluminal characteristic cones across all galactic radii.

2. **Scalar-Vector-Tensor Preferred Frame Completion:** Introducing a unit timelike vector field u^μ ($u_\mu u^\mu = -1$) and projecting gradients orthogonally ($\hat{\nabla}_\mu\phi = \nabla_\mu\phi + u_\mu u^\nu\nabla_\nu\phi$) locks perturbation propagation to the spacetime lightcone ($c_s = c$) and exactly reduces to $\mathcal{F}_{\rm dual}(x)$ in the quasistatic weak field [**P**].

3. **Disformal Metric Coupling and PPN:** Coupling matter to the disformal metric $\tilde{g}_{\mu\nu} = e^{-2\phi}g_{\mu\nu} - 2\sinh(2\phi)u_\mu u_\nu$ yields equal temporal and spatial perturbation potentials ($\tilde{h}_{ij} = -\tilde{h}_{00}$), proving $\gamma_{\rm PPN} = 1.00000$ and preferred-frame parameters $\alpha_1 = \alpha_2 = 0$ [**P**]-conditional.

4. **Cosmological Decoupling:** On spatially homogeneous FLRW backgrounds, the projected gradient vanishes identically ($\hat{\nabla}_\mu\phi(t) = 0$), so the spatial MOND kinetic term contributes zero dynamical dark energy density [**P**]. The relation $\Omega_\Lambda = \ln 2$ is strictly a horizon entropy boundary condition and remains quarantined as [**O**].

11.1 Tensor-Mode Speed vs. GW170817 Bound and Disformal Cone Analysis (Target D8 & D8b)

The characteristic speed of transverse-traceless (TT) gravitational waves has been evaluated against the multi-messenger constraint $|c_T/c_\gamma - 1| \leq 10^{-15}$ from GW170817/GRB 170817A:

1. **Einstein-Frame Vector Sector Identity:** The vector kinetic Lagrangian $-\frac{K}{32\pi G}F^{(u)}_{\mu\nu}F^{\mu\nu}_{(u)}$ uniquely forces Jacobson–Mattingly parameters $c_1 = K/2, c_3 = -K/2, c_2 = c_4 = 0$. Consequently, the tensor mode parameter $c_{13} \equiv c_1 + c_3 = 0$ is an exact algebraic identity, yielding Einstein-frame speed $c_T^2(g) = 1/(1 - c_{13}) = 1$ identically [**P**].

2. **Foster–Jacobson Preferred-Frame Bound:** The preferred-frame PPN parameter evaluates to $\alpha_1 = -2K$ [**P**]. Solar System constraints ($|\alpha_1| \lesssim 10^{-4}$) do not set $K = 0$, but place the empirical bound $|K| \lesssim 5 \times 10^{-5}$.

3. **Disformal Null Cone Split and Falsification:** Photons propagate on physical null geodesics of $\tilde{g}_{\mu\nu} = e^{-2\phi}g_{\mu\nu} - 2\sinh(2\phi)u_\mu u_\nu$, giving coordinate speed $c_\gamma = e^{2\phi}$. Gravitons on $g_{\mu\nu}$ propagate at $c_T = 1$, yielding speed ratio $c_T/c_\gamma = e^{-2\phi}$ [**P**]. This ratio equals unity if and only if $\phi = 0$. In galaxy halos ($\phi \sim 10^{-6}$) and cosmological backgrounds ($\phi \sim 0.1$), the fractional deviation $|e^{-2\phi} - 1| \sim 2|\phi| \gg 10^{-15}$ violates the GW170817 bound by orders of magnitude. The specific disformal completion with $B(\phi) = -2\sinh(2\phi)$ is therefore **FALSIFIED** [**P**], restricting viable completions to strictly conformal couplings ($B = 0$) or the Skordis–Złośnik action class.

11.2 Skordis–Złośnik Relativistic MOND Parent Membership (Target D9)

The derived dual-channel closure $\mathcal{F}_{\text{dual}}(x) = \frac{1}{2}x^2 - x + \ln(1+x)$ has been tested for exact embedding into the Skordis–Złośnik (2021) relativistic MOND framework:

1. **Covariant Action Matching:** The Skordis–Złośnik action features a free kinetic scalar function $\mathcal{J}(\mathcal{Y})$ on the spacelike projected gradient $\mathcal{Y} \equiv \frac{1}{a_0^2} g^{\mu\nu} \hat{\nabla}_\mu \phi \hat{\nabla}_\nu \phi \geq 0$. Setting
$$\mathcal{J}(\mathcal{Y}) = \frac{1}{2}\mathcal{Y} - \sqrt{\mathcal{Y}} + \ln(1+\sqrt{\mathcal{Y}}) \equiv \mathcal{F}_{\text{dual}}(\sqrt{\mathcal{Y}}) \tag{13}$$
yields the first derivative $2\mathcal{J}'(\mathcal{Y}) = \frac{\sqrt{\mathcal{Y}}}{1+\sqrt{\mathcal{Y}}} \equiv \mu(\sqrt{\mathcal{Y}})$ [**P**].

2. **Quasistatic Halo Field Equation:** In static galaxy halos ($\mathcal{Y} = x^2 = |\nabla\phi|^2/a_0^2$), the scalar Euler–Lagrange equation reduces identically to the verified dual-channel AQUAL law $\nabla \cdot [\mu(|\nabla\phi|/a_0)\nabla\phi] = 4\pi G \rho_{\text{bar}}$ [**P**].

3. **GW170817 and Lensing Concordance:** Because matter and photons couple directly to the physical Einstein metric $g_{\mu\nu}$, transverse-traceless gravitons and photons propagate on identical null cones ($c_T = c_\gamma = c$), satisfying $|c_T/c_\gamma - 1| = 0 \leq 10^{-15}$ identically [**P**]. The dynamical vector field A^μ generates an anisotropic shear stress that cancels scalar trace discrepancies, maintaining $\Phi = \Psi$ ($\gamma_{\text{PPN}} = 1$) without disformal metric distortions [**P**].

4. **Kinetic Convexity and Stability:** The second derivative $\mathcal{J}''(\mathcal{Y}) = \frac{1}{4\sqrt{\mathcal{Y}}(1+\sqrt{\mathcal{Y}})^2} > 0$ is strictly positive for all $\mathcal{Y} > 0$ [**P**], guaranteeing positive perturbation energy and freedom from ghost/gradient instabilities.

12 Limitations and Falsifiability

The proposed theoretical program is subject to clear empirical and mathematical falsification criteria:

1. **Variational Falsification:** If no covariant Lagrangian completes the step from $\mathcal{F}'(x) = \operatorname{arcsinh}(x)$ to $\mu_{\text{simple}}(x)$ without ad-hoc geometric projections, the derivation fails.

2. **Out-of-Sample Failure:** If out-of-sample rotation curves or blind kinematic surveys systematically deviate from the predicted μ_{simple} profile.

3. **Astrophysical Constraints:** Failure to satisfy Solar System ephemerides, binary pulsar orbital decay rates, gravitational lensing profiles of galaxy clusters, or cosmological perturbation spectra.

4. **Relativistic Instabilities:** Detection of ghost modes, gradient instabilities, or superluminal propagation in the covariant completion.

13 Conclusion

This technical assessment establishes that while the algebraic and formal Lean 4 implementations of the framework are mathematically sound, the complete framework relies upon several unproved physical hypotheses: the variational simple-μ closure, the a_0 normalization divisor, and the effective optical couplings. Empirical SPARC analysis confirms that the zero-parameter model outperforms fixed baryonic baselines but remains a poor absolute fit across uncurated data, while nuisance parameterization provides expected observational flexibility. No claim of a completed unified physical theory is supported by the current evidence.

Data and Code Availability

All raw data, Python fitting scripts, Lean 4 formalization proofs, and verification manifests are locally archived in the repository at `VERIFICATION_RUN_002/`.

Declarations

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Appendix A: Formal Theorem Inventory (Lean 4)

The mathematical identities associated with algebraic, geometric, and dynamical formalizations are implemented in Lean 4 and verified using Mathlib. The full formal suite of 7 modules builds cleanly with `exit code 0`. All headline theorems depend strictly on the standard foundational logical axioms:

```
[propext, Classical.choice, Quot.sound]
```

Appendix B: Full SPARC Metric Definitions and Statistics

For each individual galaxy $g \in \{1, \ldots, 175\}$ with N_g observed kinematic radii R_i, observed velocities $V_{\mathrm{obs},i}$, and observational uncertainties $\sigma_{V,i}$, the model velocity is evaluated as:

$$V_{\mathrm{model}}(R_i) = V_{\mathrm{bar}}(R_i) \cdot \left[\frac{1}{2} + \sqrt{\frac{1}{4} + \frac{a_0}{g_{\mathrm{bar}}(R_i)}} \right]^{1/2}, \tag{14}$$

where $g_{\mathrm{bar}}(R_i) = V_{\mathrm{bar}}(R_i)^2/R_i$, with $V_{\mathrm{bar}}^2(R) = |V_{\mathrm{gas}}|V_{\mathrm{gas}} + \Upsilon_{\mathrm{disk}}|V_{\mathrm{disk}}|V_{\mathrm{disk}} + \Upsilon_{\mathrm{bulge}}|V_{\mathrm{bulge}}|V_{\mathrm{bulge}}$.

Per-Galaxy Metric Definition

The per-galaxy data-residual statistic is defined as:

$$\frac{\chi^2_{\mathrm{data}}}{N_g} = \frac{1}{N_g} \sum_{i=1}^{N_g} \left(\frac{V_{\mathrm{obs},i} - f_d V_{\mathrm{model}}(f_d R_i)}{\sigma_{V,i}} \right)^2. \tag{15}$$

Aggregate Data-Residual Metric Definition

The aggregate data-residual statistic across the complete sample of $N_{\mathrm{data}} = 3,391$ observations is:

$$\frac{\sum \chi^2_{\mathrm{data}}}{\mathrm{DOF}_{\mathrm{nom}}} = \frac{1}{N_{\mathrm{data}} - N_{\mathrm{fitted}}} \sum_{g=1}^{175} \sum_{i=1}^{N_g} \left(\frac{V_{\mathrm{obs},i} - f_d V_{\mathrm{model}}(f_d R_i)}{\sigma_{V,i}} \right)^2. \tag{16}$$

Appendix C: Nuisance Parameter Derivation, Nominal DOF, and Cross-Validation

Across the full SPARC database ($N = 175$ galaxies, $N_{\mathrm{data}} = 3,391$ data points):

- **32 Galaxies with Active Bulges ($V_{\mathbf{bulge}} > 0.5$ km/s):** 3 fitted parameters each ($\Upsilon_{\mathrm{disk}}, \Upsilon_{\mathrm{bulge}}, f_d$) $\implies$ $32 \times 3 = 96$ parameters.
- **143 Galaxies without Active Bulges ($V_{\mathbf{bulge}} \leq 0.5$ km/s):** 2 fitted parameters each ($\Upsilon_{\mathrm{disk}}, f_d$) $\implies$ $143 \times 2 = 286$ parameters.

$$N_{\mathrm{par}} = 96 + 286 = 382 \text{ fitted parameters}, \qquad \mathrm{DOF}_{\mathrm{nom}} = 3,391 - 382 = 3,009 \text{ nominal dof.} \tag{17}$$

Lean 4 Module	Headline Theorems	Formal Proposition Verified
`SOCasimirGenuine.lean`	`casimir_defining_rep`	Quadratic Casimir eigenvalue of standard $\mathfrak{so}(n)$ generators is $(n-1)/2$.
`DeSitterExtremal.lean`	`desitter_lapse_horizon`	Lapse $1 - H^2r^2 = 0$ at $r = 1/H$, and arithmetic positivity of $cH/(2\pi)$.
`MuProjection.lean`	`mu_simple_eq_cos`, `powerLaw_iterated_deriv`	Right-triangle projection properties of $\mu_{\text{simple}}(x)$ and iterated derivatives.
`ITActionClosure.lean`	`tauLaw_eq_simple_mu_poly`, `btfr_deep_mond`	Equivalence of τ-relation and simple-μ polynomial; BTFR $M \propto v^4$.
`YettParadigm.lean`	`ramanujan_yett_spectral_gap_pos`	Positivity of the spectral gap $\lambda_1 - \lambda_0 > 0$ for $\kappa > 0$.
`SovereignRegularity.lean`	`sovereign_regularity_theorem`	Conditional Beale–Kato–Majda integral boundedness under the hypothesis $\chi \geq \theta$.
`GODActionKinematics.lean`	`dual_channel_poly_identity`, `btfr_algebraic_scaling`	Exact dual-channel quadratic identity and point-mass BTFR algebraic scaling.

Table 2: Inventory of formally verified Lean 4 theorems in `05_lean_formalization/`.

5-Fold Cross-Validation Protocol

To evaluate out-of-sample generalization and eliminate overfitting artifacts, the 175 galaxies were partitioned into 5 independent folds ($k = 5$, 35 galaxies per fold). Evaluating each test fold against fixed population priors ($\Upsilon_{\text{disk}} = 0.5, \Upsilon_{\text{bulge}} = 0.7, f_d = 1.0, a_0 = 1.2 \times 10^{-10}$ m/s^2) without per-galaxy optimization yields an out-of-sample median $\chi^2_{\text{data}}/N_g = 14.68$ and aggregate $\chi^2_{\text{data}}/\text{dof} =$

110.18, outperforming fixed unit-M/L baselines while demonstrating true predictive bounds.

Appendix D: Reproduction Commands and Software Environment

```
# 1. Verify all Lean 4 formalization modules
cd /home/mega/Chyren/Research_and_Data/03_Formal_and_Lean/formal
lake env lean /home/mega/grand_monograph/05_lean_formalization/GODActionKinematics.lean

# 2. Run SPARC 175-galaxy benchmark & 5-Fold Cross Validation
cd /home/mega/grand_monograph/02_galaxy_dynamics
python3 sparc_cross_validation.py

# 3. Compile final manuscript PDF
cd /home/mega/grand_monograph
pdflatex -interaction=nonstopmode final_manuscript.tex
bibtex final_manuscript
pdflatex -interaction=nonstopmode final_manuscript.tex
pdflatex -interaction=nonstopmode final_manuscript.tex
```

Appendix E: Artifact Manifest and SHA-256 Table

Artifact File	**Description**	**SHA-256 Checksum**
`sparc_raw_inputs`	175 raw SPARC data files	`e76e6752164b...c05f96d095456e067bdd5c6da59d2be4ec70c1eb`
`sparc_cross_validation.py`	Python cross-validation	`a4b12c88414a...971842617f69201cb4d1276a0891461f32a76f21`
`SPARC_CV_REPORT.json`	5-Fold Cross Validation	`c8430e79124a...bf134988771142567990145266491742416f491c`
`LEAN_BUILD_RAW.json`	Raw Lean 4 build report	`1ea16027c92b...a079c94626127e26bf484252676766432f4a434c`

Table 3: SHA-256 checksums of core reproducibility artifacts.